National University of Computer and Emerging Sciences
Karachi Campus

Programming for AI (AI-2001)

Sessional-II Exam

Total Time (hrs): 01
Total Marks: 50
Total Questions: 02

Date: **November 06, 2025**
Time: **01:45 PM – 02:45 PM**

Course Instructor
Mr. Talha Shahid

Student Name	Roll No	Section	Student Signature
		BAI-3A	

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CLO # 2: Understand and apply the **Object-Oriented** concepts in the python.

Q1: **[marks: 20]**

Talha Shahid

You are a junior developer at **`TechServe`**, a company that sells high-end electronics. The company needs to upgrade its inventory system to better model its complex products.

A new requirement has come in: products must be represented by combining two separate concepts:

- **Core Product Data:** Basic information like **`ID`** and **`name`**.
- **Pricing Data:** Complex pricing information, including a base price and a tax rate.

Your task is to create a set of classes to model a **`ServiceableProduct`**, which has core **`data`**, **`pricing data`**, and its own service history.

1. Create the `Product` Class (5 Points)
- **(2 pts)** Create a class named **`Product`**. It must accept **`productId`** and **`name`**.
- **(3 pts)** Create a method named **`getProductInfo`**. This method must **`return`** a single formatted string: e.g., **`"ID: productId | Name: name"`**

2. Create the `Pricing` Class (5 Points)
- **(2 pts)** Create a class named **`Pricing`**. It must accept **`basePrice`** and **`taxRate`** (e.g., **`0.08`** for 8%).
- **(3 pts)** Create a method named **`getFinalPrice`**. This method must **calculate** the final price (**`basePrice * (1 + taxRate)`**) and **`return`** the result.

3. Create the `ServiceableProduct` Class (7 Points)
- **(3 pts)** Create a class named **`ServiceableProduct`** that inherits from **both `Product`** and **`Pricing`**.
  - It must accept all necessary arguments to initialize its parents and itself:
  - A **list of dictionaries** is to be created name with **`serviceLog`**.
  - You must ensure the **`__init__`** methods of parent classes are called correctly.

- **(2 pts)** Create a method **getServiceHistory**.
  - This method should iterate through the **serviceLog** (a list of dictionaries like **{'date': '...', 'issue': '...'}**).
  - It must extract **only the 'issue' string** from each dictionary.
  - It must **return** a single string of all issues, joined by **", "**, and formatted like this: **"Maintenance Log: [issue1, issue2]"**
- **(2 pts)** Create a method **getFullDetails**.
  - This method must *call* **getProductInfo** from its parent and **getFinalPrice** from its *other* parent.
  - It must **return** a single-line string combining these results, formatted exactly as: **"Product: [info_from_get_product_info] --- Final Price: $[price_from_get_final_price]"**
  - **Note:** This method should *not* call **getServiceHistory**.

4. Execution and Demonstration (3 Points)

- **(1 pt)** Create one instance of **ServiceableProduct** for a **"Pro Laptop"** with the following data:
  - **ID: "P900"**
  - **Name: "Pro Laptop"**
  - **Base Price: 1500**
  - **Tax Rate: 0.10 (i.e., 10%)**
  - **Service Log: [{'date': '2024-01-15', 'issue': 'Screen repair'}, {'date': '2025-03-10', 'issue': 'Battery replacement'}]**
- **(2 pts)** Call **all three** methods from your **ServiceableProduct** object (**getProductInfo**, **getFullDetails**, **getServiceHistory**) and **print** the value returned by each call.

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Expected Output

```
ServiceableProduct __init__ called
BaseProduct __init__ called
Pricing __init__ called

---- Demonstration ----

Calling get_product_info():
ID: P900 | Name: Pro Laptop

Calling get_full_details():
Product: [ID: P900 | Name: Pro Laptop] --- Final Price: PKR 1650.00

Calling get_service_history():
Maintenance Log: [Screen repair, Battery replacement]
```

CLO # 3: Analyze and solve problems using programming, **data analysis** and **standard libraries and/or toolboxes** of the programming language.

Q2: **[marks: 30]**

You are a junior data analyst at a telecom company. You have been given a small dataset of customer information. Your goal is to **clean the data**, engineer **new features**, perform **advanced analysis**, and create a clear **visualization**.
Note: Churn means whether a customer has left the service (`1` = left, `0` = stayed).

DataFrame: Assume you have loaded the following data into a `DataFrame`.

	CustomerID	Gender	Age	Contract	MonthlyBill	Churn
0	101	Male	25.0	Monthly	50	0
1	102	Female	45.0	Annual	100	1
2	103	Female	30.0	Monthly	45	0
3	104	Male	50.0	Annual	110	1
4	105	Female	22.0	Monthly	55	0
5	106	None	35.0	Monthly	60	0
6	107	Male	NaN	Annual	105	1
7	108	Female	60.0	None	95	0
8	109	Male	42.0	Two Year	120	1
9	110	Female	33.0	Monthly	65	0

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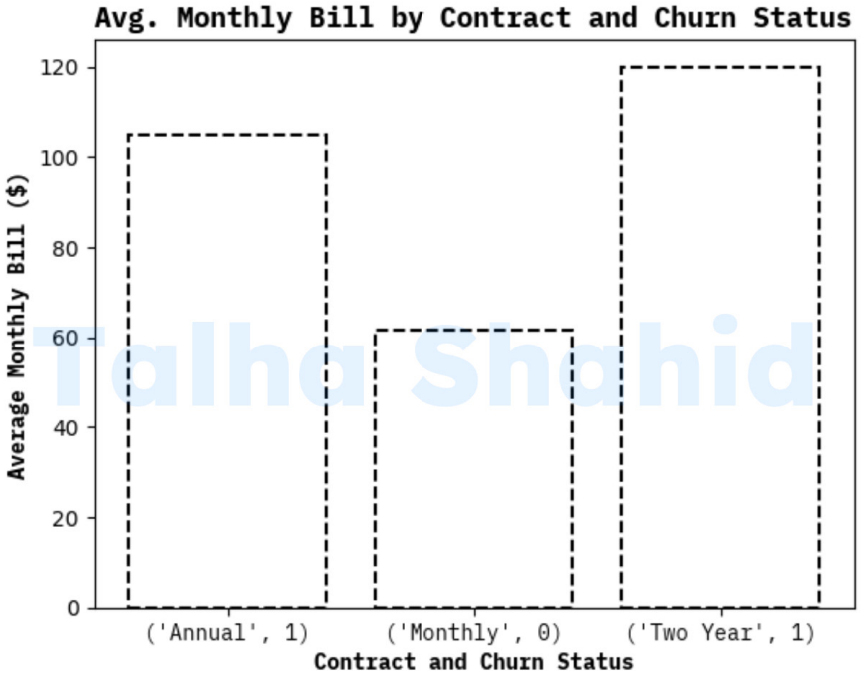
Write the Python code to perform the following steps in order:

1. **Setup:** Import modules
2. **Data Cleaning (3 pts):**
   - Fill the missing values.
3. **Feature Engineering (12 pts):**
   - Create a new column named `AgeGroup`. Bin the Age column into three equal-width bins labeled: `'Young'`, `'Adult'`, `'Senior'`.
   - Create a new column `HighRisk` which is `1` if the Contract is `'Monthly'` and `MonthlyBill` is greater than `100`, and `0` otherwise.
4. **Analysis (6 pts):**
   - Create and print a summary table that groups the data by `Contract` *and* `AgeGroup`. This table must show the **count** of customers and the **mean** `MonthlyBill` for each sub-group.
5. **Visualization (9 pts):**
   - First, create a Pandas Series named `avGBills` that contains the average `MonthlyBill`, grouped by both `Contract` and `Churn`.
   - Using **Matplotlib's** object-oriented (OOP) approach:
     - Create a `fig` and ax object.
     - Prepare the `labels` (as strings) and `values` from the `avgBills` Series.
     - Create a `bar chart` on the `ax` object with `labels`, `values`.
     - Set the title to: `"Avg. Monthly Bill by Contract and Churn Status"`
     - Set the **y-axis** label to: `"Average Monthly Bill ($)"`
     - Set the **x-axis** label to: `"Contract and Churn Status"`
     - Show the plot.

Expected Output

```
            --- Analysis Summary Table ---
                      Customer_Count  Mean_Monthly_Bill
Contract  AgeGroup
Annual    Adult                    2             102.50
          Senior                   1             110.00
Monthly   Young                    4              53.75
          Adult                    1              60.00
          Senior                   1              95.00
Two Year  Adult                    1             120.00
```

```
--- Plotting Code (output will be a graph) ---
```



Talha Shahid